

Single-phase AC Circuit

AC Circuit with series Elements :-

Interconnect by considering two elements
R-L, R-C, LC and connection of three elements R-L-C.

1) Series R-L Circuit :-

The AC voltage source of instantaneous voltage
 $v = V_m \sin \omega t$
connected a/c a combination
of R & L.

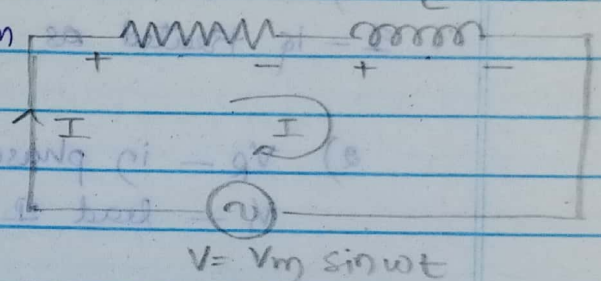
R - ohm

L - Henry

Current i in L & R is I amp.

I - rms value

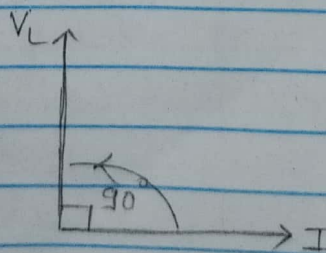
i - instantaneous value



Due to the voltage drop across element L & R is -

$$V_R = I \cdot R \quad (V_R \text{ \& } I \text{ are in phase})$$

$$V_L = I \cdot X_L \quad (V_L \text{ lead } I \text{ by } 90^\circ)$$



2, 3, 9, 18, 20
27, 28
22, 34, 40
3, 3, 1

Phasor diagram :-

Applying KVL to the diagram -

$$\bar{V} - \bar{V}_R - \bar{V}_L = 0$$

$$\bar{V} = \bar{V}_R + \bar{V}_L$$

$$\bar{V} = \bar{I}R + \bar{I}X_L$$

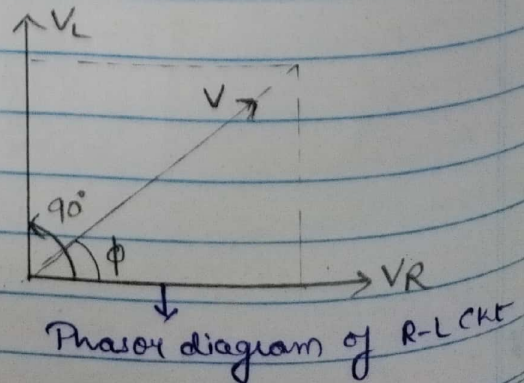
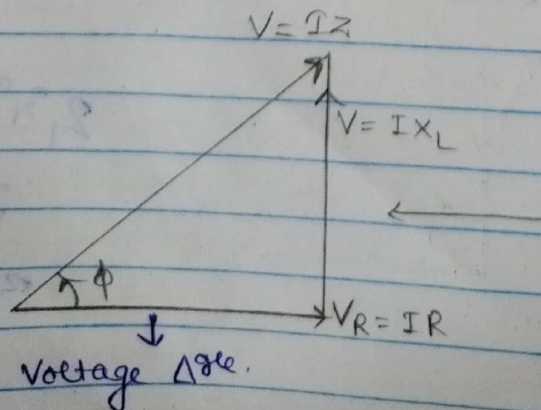
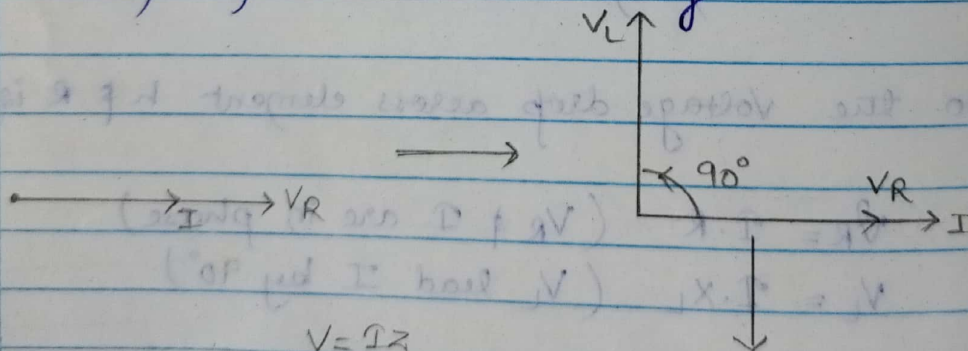
Steps to be followed to draw the phasor diagram -

1) As one current flowing in both R & L is same,
 $\therefore I$ - is treated as reference phasor

2) V_R - in phase with I
 V_L - lead I by 90°

3) Draw the resultant of one circuit which is the supply voltage

4) Hence draw the Voltage Δ gle.



Vector Diagram —

$$\vec{V} = \vec{V}_R + \vec{V}_L$$

$$\vec{V} = \vec{I}R + \vec{I}X_L$$

$$V = I \sqrt{R^2 + X_L^2}$$

$$\sqrt{R^2 + X_L^2} = |Z|$$

$$\underline{IZ} = \underline{IR} + \underline{IX}_L$$

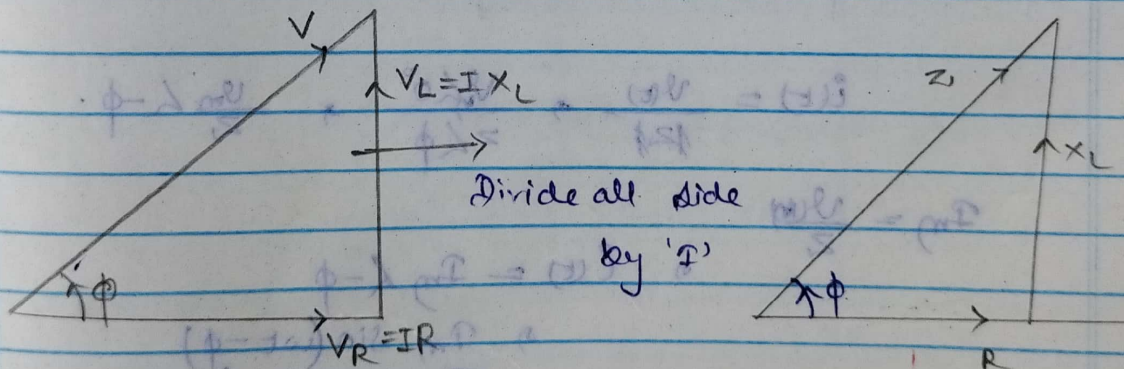
∴ Impedance is given as —

$$Z = R + jX_L$$

$$Z = |Z| \angle \phi$$

$$|Z| = \sqrt{R^2 + X_L^2}$$

Impedance Angle :-



$$Z = |Z| \angle \phi$$

$$Z = \sqrt{R^2 + X_L^2} \quad \phi = \tan^{-1} \phi$$

Impedance Angle.

X_L — inductive reactance

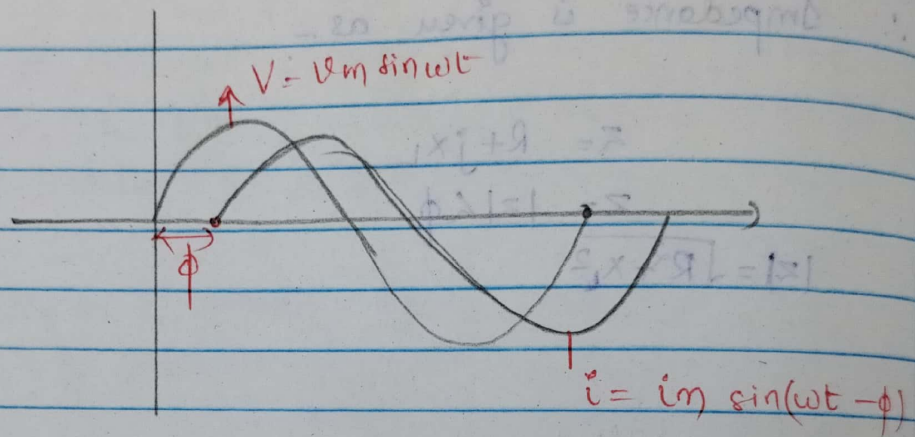
$$X_L = 2\pi fL$$

Phase relation:-

$$v = V_m \sin \omega t$$

$$i = I_m \sin(\omega t - \phi)$$

v leads i by angle ϕ in R-L circuit



Expression for current-

$$i(t) = \frac{v(t)}{Z} \Rightarrow \frac{V_m \angle 0^\circ}{Z \angle \phi} \Rightarrow \frac{V_m}{Z} \angle -\phi$$

$$I_m = \frac{V_m}{Z}$$

$$i(t) = I_m \angle -\phi$$

$$\underline{i(t) = I_m \sin(\omega t - \phi)}$$

$$i = I_m \sin \omega t \Rightarrow I_m \angle 0^\circ \text{ --- polar.}$$

$$v = Z(R + jX_L)$$

$$v(t) = i(t) \sqrt{R^2 + X_L^2}$$

↓

$$i(t) = \frac{v(t)}{Z} \Rightarrow \frac{V_m \angle 0^\circ}{Z \angle \phi}$$

Power in L-R Series ckt -

$$P = V \times i$$

$$= V_m \sin \omega t \times i \sin (\omega t - \phi)$$

$$= V_m I_m [\sin \omega t \sin (\omega t - \phi)]$$

$$2 \sin A \sin B =$$

$$\sin(A+B) \cos$$

$$\Rightarrow \cos(A-B) - \cos(A+B)$$

$$V = IR$$

$$I =$$

$$V =$$

$$P = \frac{V_m I_m}{2} [\cos(\phi) - \cos(2\omega t - \phi)]$$

$$P = \frac{V_m I_m}{2} \cos \phi - \frac{V_m I_m}{2} \cos(2\omega t - \phi)$$

Avg. power -

$$P_{avg} = \frac{1}{2\pi} \int_0^{2\pi} \frac{V_m I_m}{2} \cos \phi d\omega t - \frac{1}{2\pi} \int_0^{2\pi} \frac{V_m I_m}{2} \cos(2\omega t - \phi) d\omega t$$

$$P_{avg} = \frac{1}{2\pi} \int_0^{2\pi}$$

Avg. power

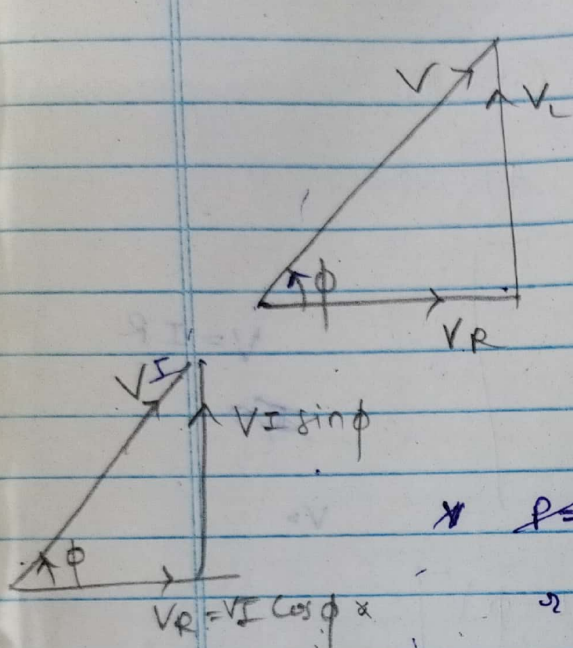
Over complete cycle = 0

Over complete

$$P_{avg} = \frac{V_m I_m}{2} \cos \phi$$

$$P_{avg} = V_{rms} \times I_{rms} \times \cos \phi \text{ watts}$$

Power Angle $\rightarrow$



voltage
power angle

$$V = V_L + V_R \quad \text{---} \times I$$

$$\underline{VI} = \underline{I V_L} + \underline{I V_R} \quad \text{---} \rightarrow$$

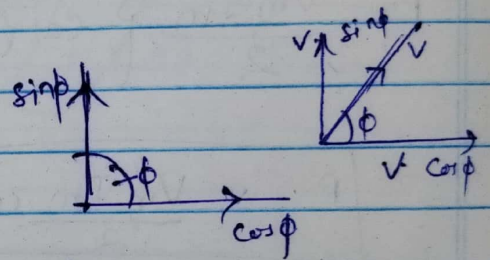
$$P_{\text{total}} = \downarrow \quad \text{---} \rightarrow \text{resist.}$$

$$\quad \quad \quad \text{Induct.}$$

$$IV = VI \cos \phi + VI \sin \phi$$

$$P = I^2 X_L + I^2 R$$

Important terms —



1) Apparent power —

$$\text{App} (s) = V \times I \quad \text{---} \rightarrow$$

$$\quad \quad \quad \downarrow \quad \quad \quad \text{---} \rightarrow \text{Rms Current}$$

$$\quad \quad \quad \text{Rms Voltage}$$

Unit (V.A) Volt-Amp.

Kilo Volt-Amp.

2) True power / Active power / Real power

Ang. power taken or consumed by or
ckt

$$P = P_{\text{av}} = VI \cos \phi \quad \text{--- unit watts.}$$

$$\phi \text{ angle b/w } V \text{ \& } I.$$

or real power is the power consumed by one resistor
in one ckt.

3) Reactive power

Reactive power — imaginary power — responsible for losses.

$$Q = VI \sin \phi \quad \text{VAR}$$

$$\propto \text{KVAR}$$

Power factor —

Ratio of True power / total (Apparent power)

$$\text{P.F.} = \frac{\text{True power}}{\text{Apparent power}} = \frac{VI \cos \phi}{VI} = \cos \phi$$

Power factor can also be defined as one ratio of the impedances;

$$\text{ie } R = |Z| \cos \phi$$

$$Z = R + jX_L \Rightarrow Z \cos \phi + j Z \sin \phi$$

$$R = Z \cos \phi$$

$$\cos \phi = \left(\frac{R}{Z} \right)$$

$$-1 \leq \text{P.F.} \leq 1$$

P.F. , leading lagging Unit.

i lead.

∴ cos φ — +ve

Cap.

i lag →

∴ cos φ — -ve

Ind.

→ V & i — in phase
∴ cos φ = 1

Resis.

Q. Expr. for current when voltage is -

$$v = 283 \sin 100\pi t$$

is applied to a coil having

$$R = 100\Omega \quad L = 0.318H$$

Solⁿ - $v = V_m \sin \omega t$

$$V_m = 283V \quad \omega = 100\pi = 2\pi f \text{ rad/sec.}$$

$$f = 50Hz$$

$$X_L = 2\pi fL$$

$$i = I_m \sin(\omega t - \phi)$$

$$= 100 \times 0.318 = \frac{31.8}{1000} = 31.8$$

$$I_m = \frac{V_m}{Z} = \frac{283}{141.42} = 2.0013 \text{ Amp.}$$

$$Z = \sqrt{R^2 + X_L^2}$$

$$= \sqrt{100^2 + 100^2}$$

$$= 100\sqrt{2}$$

$$= 141.42$$

$$\phi = \tan^{-1} \left(\frac{X_L}{R} \right)$$

$$\phi = 45^\circ$$

$$i = 2 \sin(100\pi t - 45^\circ)$$

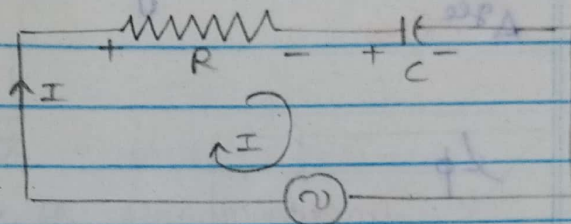
Series R-C Circuit :-

A pure capacitor of value 'C' connected in series with a resistor 'R' and a voltage source of instantaneous voltage, $v = V_m \sin \omega t$, is connected across a series combination of R-C.

$$v = V_m \sin \omega t$$

$$V_R = IR$$

$$V_C = IX_C$$

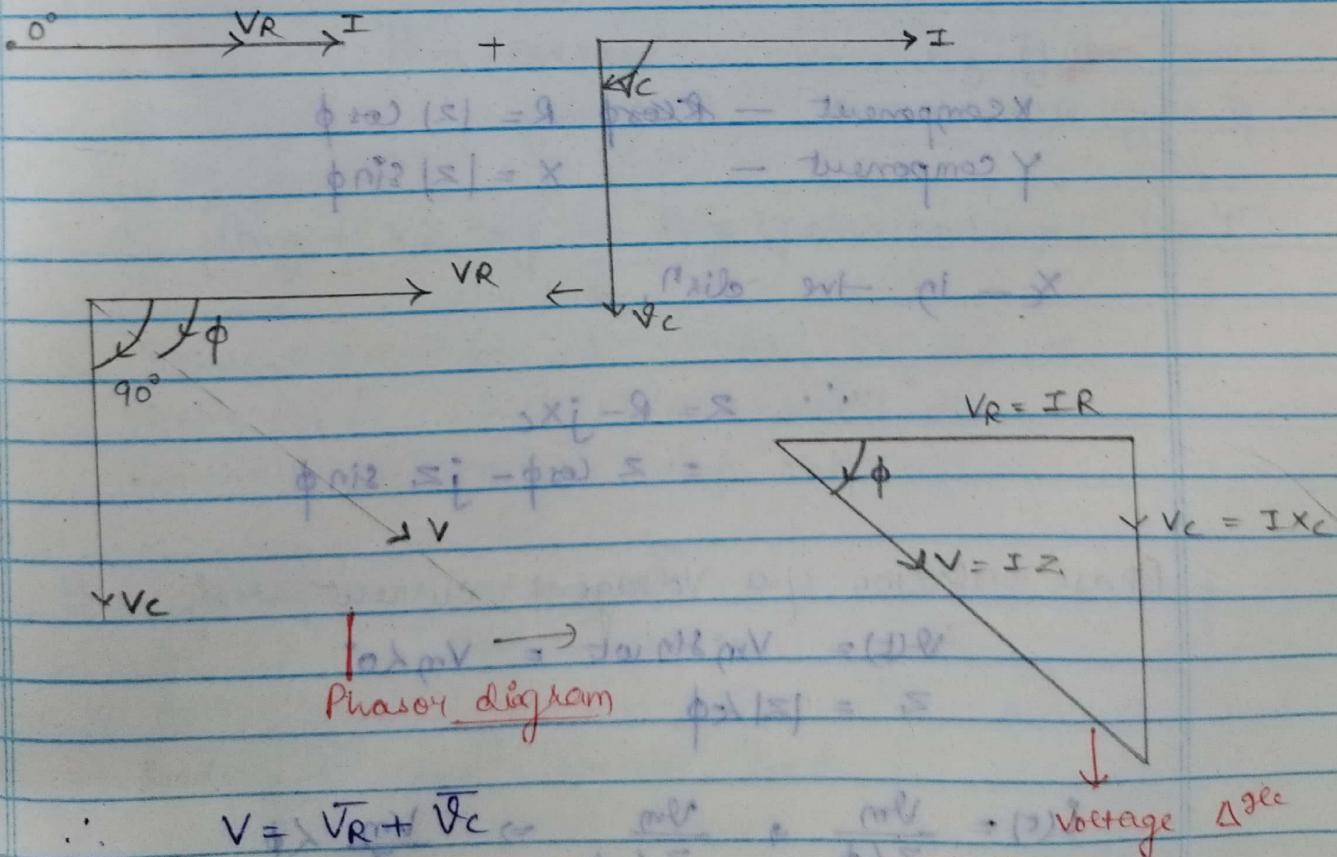


$$v = V_m \sin \omega t$$

$V_R \rightarrow I \rightarrow$ in phase $\phi = 0^\circ$

$V_C \rightarrow I \rightarrow$ lead V_C by $\phi = 90^\circ$.

Phasor diagram $\rightarrow$

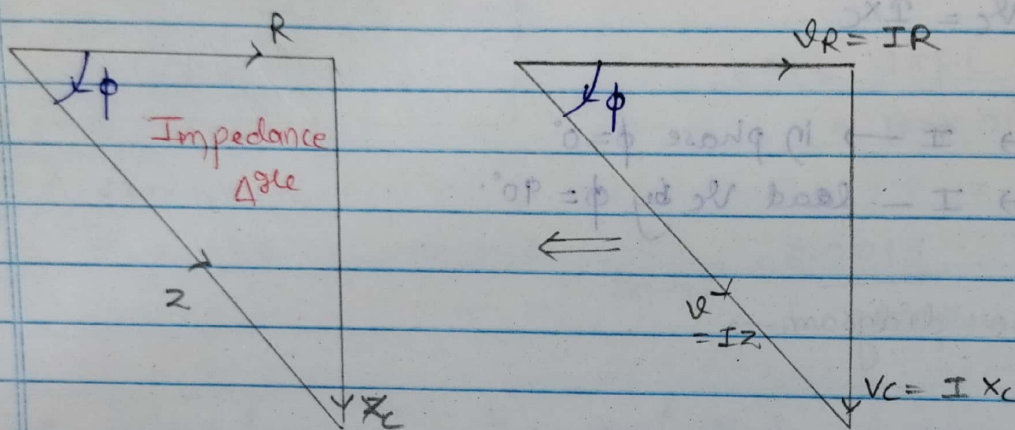


$$Z = R - jX_C \text{ or } R + \frac{1}{jX_C}$$

$$Z = \sqrt{R^2 + X_C^2} \quad \phi = \tan^{-1}\left(\frac{-X_C}{R}\right)$$

Impedance Δ_{ϕ} —

Δ_{ϕ} Voltage Δ_{ϕ} can be converted to one in



$$X \text{ component} - R \cos \phi \quad R = |Z| \cos \phi$$

$$Y \text{ component} - \quad X = |Z| \sin \phi$$

X_C — in —ve dirⁿ,

$$\therefore Z = R - jX_C \\ = Z \cos \phi - jZ \sin \phi$$

Phase relation b/w Voltage & current —

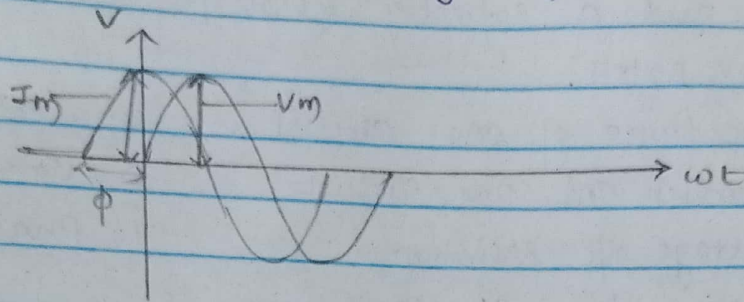
$$V(t) = V_m \sin \omega t = V_m \angle 0^\circ$$

$$Z = |Z| \angle -\phi$$

$$i(t) = \frac{V_m}{Z \angle -\phi} \rightarrow \frac{V_m}{Z} \angle \phi$$

$$i(t) = I_m \sin(\omega t + \phi)$$

ie current leads voltage by angle ' ϕ '



Power in RC circuit :-

$$p = v \times i$$

$$p = V_m \sin \omega t \times I_m \sin(\omega t + \phi)$$

$$= V_m I_m [\sin \omega t \times \sin(\omega t + \phi)]$$

$$P = \frac{V_m I_m \cos \phi}{2} - \frac{V_m I_m \cos(2\omega t + \phi)}{2}$$

↓
P_{avg} (Average power)

↓
Avg. of one power over complete cycle is '0'

$$\therefore \boxed{P_{avg} = VI \cos \phi} \text{ --- Avg. power consumed by 'R'}$$

$$\therefore \text{Apparent Power (S)} = VI \text{ (VA)}$$

$$\text{Active Power (P)} = VI \cos \phi \text{ watts}$$

$$\text{Reactive Power (Q)} = VI \sin \phi \text{ VAR}$$

$$\text{Power factor} = \frac{\text{True power}}{\text{Apparent power}} = \frac{VI \cos \phi}{VI} = \cos \phi$$

leading P.F for capacitive load.

Q/- For a series R-C circuit consisting of a resistor of 50Ω and a capacitor of $100\mu F$. Calculate, if supply is $230V, 50Hz$

1. Impedance of the circuit.
2. Current thr. the circuit
3. Voltage a/c resistor
4. Voltage a/c capacitor
5. Power factor
6. Power
7. Phasor diagram.

Soln

$$\omega = 2\pi f = 100\pi \quad V_s = 230 \text{ (RMS)}$$

$$V_m = 230 \times \sqrt{2} = 325.26 \text{ V}$$

$$V = V_m \sin \omega t = 325.26 \sin 100\pi t.$$

$$Z = R - jX_c$$

$$R = 50\Omega$$

$$50 - j100$$

$$X_c = \frac{1}{2\pi fC} = \frac{1}{2\pi \times 50 \times 100 \times 10^{-6}} = 31.847$$

$$Z = \sqrt{(50)^2 + (100)^2}$$

$$\Rightarrow 111.802$$

$$\phi = \tan^{-1} \left(\frac{-100}{50} \right) = \tan^{-1} (-2) = -63.43^\circ$$

$$Z = 111.80 \angle -63.43^\circ$$

$$Z = R - jX_c \Rightarrow$$

$$50 - j31.847$$

$$\Rightarrow 59.2802$$

$$\phi = \tan^{-1} \left(\frac{-31.847}{50} \right)$$

$$= -32.49^\circ$$

$$Z = |Z| \angle \phi = 59.280 \angle -32.49^\circ$$

$$\text{Current} = \frac{V}{Z} = \frac{230}{59.28 \angle -32.49^\circ} = 3.879 \angle 32.49^\circ \text{ Amp.}$$

$$\text{Voltage a/c resistor, } V = IR = 3.879 \times 50$$

$$\Rightarrow 193.99 \approx 194 \text{ V}$$

Voltage across capacitor -

$$\begin{aligned} V &= I \times X_C \\ &= 3.879 \times 31.847 \\ &= \underline{123.56 \text{ V}} \end{aligned}$$

$$\text{Power factor} = \cos \phi = \frac{R}{Z} = \frac{50}{59.280} = \underline{0.8435 \text{ leading.}}$$

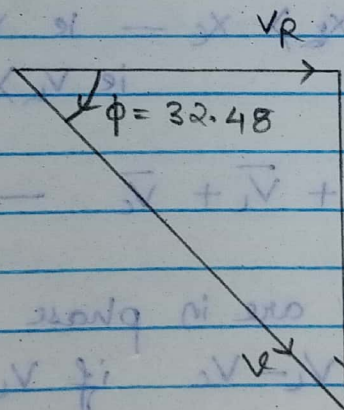
or -

$$\cos(-32.48) = \underline{0.8435 \text{ leading.}}$$

Power - True power -

$$\begin{aligned} P &= VI \cos \phi \\ &= 230 \times 3.879 \times 0.8435 \\ &= \underline{752.54 \text{ Watts.}} \end{aligned}$$

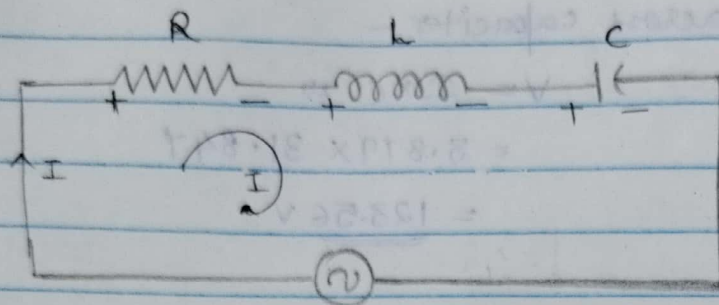
Phasor diagram -



R-L-C Series Circuit :-

In this R-L-C are connected in series with the source voltage.

$$\begin{aligned} V_R &= IR & V \text{ \& } I \text{ in phase} \\ V_L &= IX_L & V \text{ leads } I \text{ by } 90^\circ \\ V_C &= IX_C & I \text{ leads } V \text{ by } 90^\circ. \end{aligned}$$



$$v = V_m \sin \omega t$$

To draw one corresponding phaser diagram of the circuit three possible conditions arise -

$$\text{Condition 1} - X_L > X_C$$

$$\text{Condition 2} - X_L < X_C$$

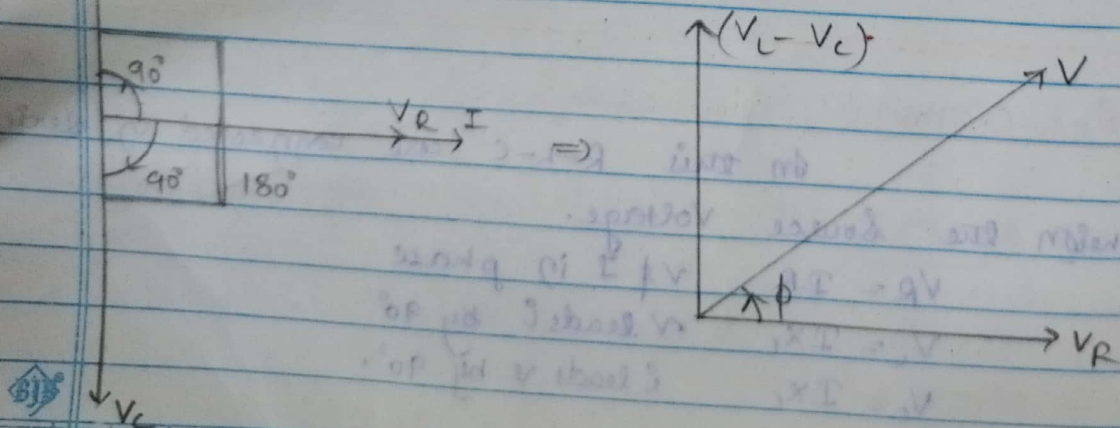
$$\text{Condition 3} - X_L = X_C$$

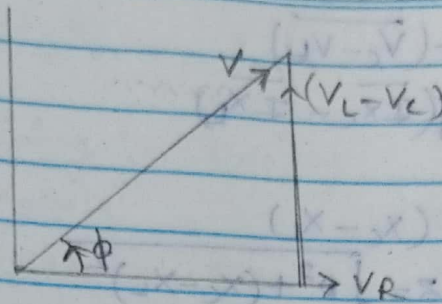
$$\text{Cond}^n(1) \Rightarrow X_L > X_C$$

If $X_L > X_C$ - ie voltage across X_L is V_L > V_C
ie $V_L > V_C$

$$V = \bar{V}_R + \bar{V}_L + \bar{V}_C \quad \text{--- phasor sum.}$$

and V_L & V_C are in phase opposition with each other
 $\therefore V_L - V_C$ if $V_L > V_C$





$V = \text{vector sum} \Rightarrow \sqrt{V_R^2 + (V_L - V_C)^2}$
 $\Rightarrow I(R + j(X_L - X_C))$

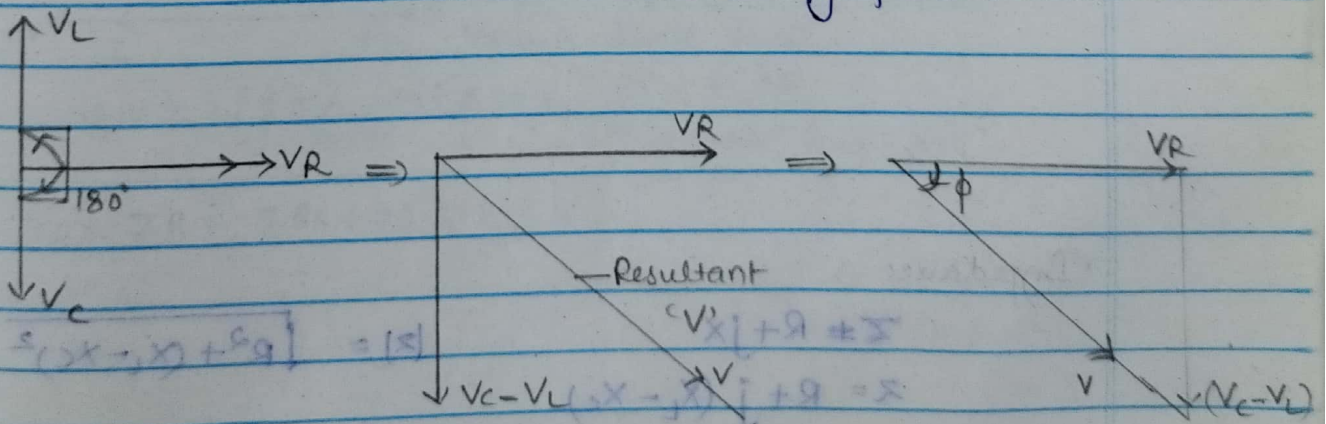
$$Z = R + j(X_L - X_C) \Rightarrow \sqrt{R^2 + (X_L - X_C)^2}$$

$$\phi = \tan^{-1} \left(\frac{X_L - X_C}{R} \right)$$

If $V_L - V_C$ i.e. $V_L > V_C$, then circuit is inductive & current lags behind voltage 'V' by ϕ .

Condition II $\rightarrow$ where $V_L < V_C$

If $V_L < V_C$ then $V_C - V_L$ or $-(V_L - V_C)$
 $\downarrow$
 180° out of phase.



$$V = \bar{V}_R + (\bar{V}_C - \bar{V}_L)$$

$$\Rightarrow \bar{I}R + j(\bar{I}X_C - \bar{I}X_L)$$

$$Z = R + j(X_C - X_L)$$

$$Z = \sqrt{R^2 + (X_C - X_L)^2}$$

$$\phi = \tan^{-1} \left(\frac{X_C - X_L}{R} \right)$$

If $V_C > V_L$ — then circuit is capacitive, current leading voltage by ϕ°

Condition - III $X_L = X_C$

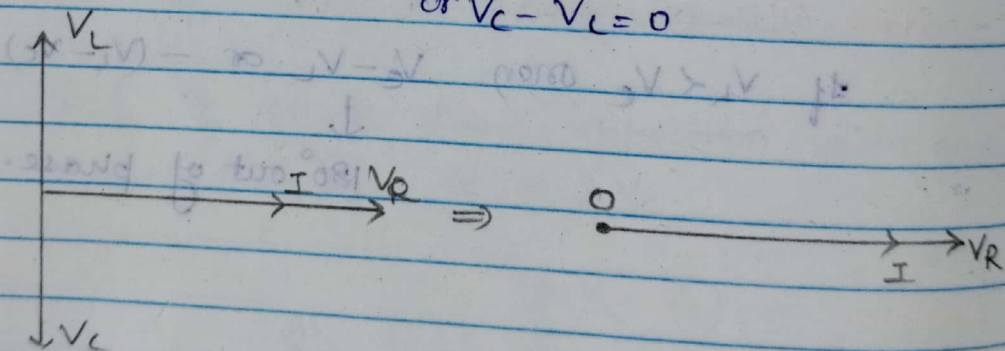
If $X_L = X_C$ — $V_L = V_C$ & they are 180° out of phase with each other

Hence, the resultant is — $V = V_R$

As $V_L = V_C = \text{same}$

$$\therefore V_L - V_C = 0$$

$$\text{or } V_C - V_L = 0$$



Impedance —

$$Z = R + jX$$

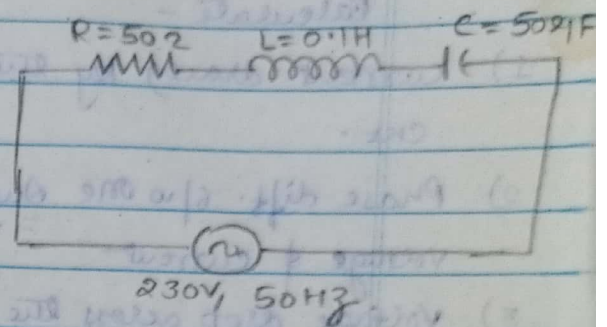
$$Z = R + j(X_L - X_C)$$

$$|Z| = \sqrt{R^2 + (X_L - X_C)^2}$$

$$\phi = \tan^{-1} \left(\frac{X_L - X_C}{R} \right)$$

Ques- for the Series RLC,
calculate

1. Current flowing thr. the ckt
2. P.F & its nature
3. Power consumed by one ckt
4. Draw the phasor diagram.



Ans- $X_L = 2\pi fL$

$$= 150\pi \times 0.1 \text{ H}$$

$$= 31.4 \Omega$$

$$X_C = \frac{1}{2\pi fC} = \frac{1}{100\pi \times 50 \times 10^{-6}}$$

$$= 63.69 \Omega$$

$$X_C > X_L$$

$$Z = \sqrt{R^2 + (X_C - X_L)^2} = 59.52$$

$$= \sqrt{50^2 + (63.69 - 31.4)^2} = 59.52 \Omega$$

$$i = \frac{V}{Z} = \frac{230}{59.52} = 3.86 \text{ Amp}$$

$$\phi = \tan^{-1} \left(\frac{X_C - X_L}{R} \right)$$

$$\rightarrow i = \frac{V}{Z} = \frac{230}{59.52} = 3.86 \text{ Amp} \quad \approx 32.85^\circ$$

$$\rightarrow \text{P.F} = \frac{R}{Z} = \frac{50}{59.52} = 0.84 \text{ leading}$$

Power consumed by the ckt -

$$P = V_{\text{rms}} \times I_{\text{rms}} \cos \phi$$

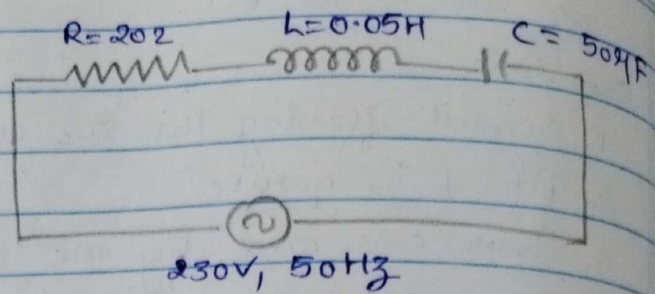
$$= 230 \times 3.86 \times 0.84$$

$$= 746.75 \text{ Watts}$$

$$V_R = IR = 3.86 \times 50 = 193 \text{ V}$$

Calculate -

- 1) current drawn by the ckt.
- 2) Phase diff. b/w one supply voltage & current
- 3) Voltage drop across the coil & Capacitor.



$$\begin{aligned} \text{Sol}^n \rightarrow X_L &\rightarrow 2\pi fL & X_C &\rightarrow \frac{1}{2\pi fC} \rightarrow \frac{1}{100\pi \times 50 \times 10^{-6}} \\ &= 100\pi \times 0.05 & & \\ R &= 20\Omega & & \\ &= 15.7\Omega & & \rightarrow 63.69\Omega \end{aligned}$$

$$i = \frac{V}{Z} = \frac{230}{\sqrt{R^2 + (X_C - X_L)^2}} = \frac{230}{69.30} = 3.318A$$

$$Z = \sqrt{(20)^2 + (63.69 - 15.7)^2} = 51.99 \rightarrow \frac{230}{51.99} = 4.42A$$

$$\begin{aligned} \phi &= \tan^{-1} \frac{-(X_C - X_L)}{R} = \tan^{-1} \frac{-(63.69 - 15.7)}{20} \\ &= \tan^{-1} \frac{-47.99}{20} = \tan^{-1} (-2.3995) = -67.37^\circ \end{aligned}$$

$$i = \frac{V}{Z \angle \phi} = \frac{230}{51.99 \angle -67.37^\circ} = 4.42 \angle 67.37^\circ \text{ Amp.}$$

$$\begin{aligned} V_L &= IX_L \\ &= 4.42 \times 15.7 \\ &= 69.30V \end{aligned}$$

Case	Nature of CRT	P.F
1) $X_L > X_C$	Inductive (R-L Series)	lagging P.F, < 1
2) $X_C > X_L$	Capacitive (R-C Series)	leading P.F, < 1
3) $X_C = X_L$	Purely resistive	$= 1$

Resonance in R-L-C Series Circuit:-

The resonance is a phenomenon taking place in the series and/or parallel combination of R-L-C circuit, which results in U.P.F.

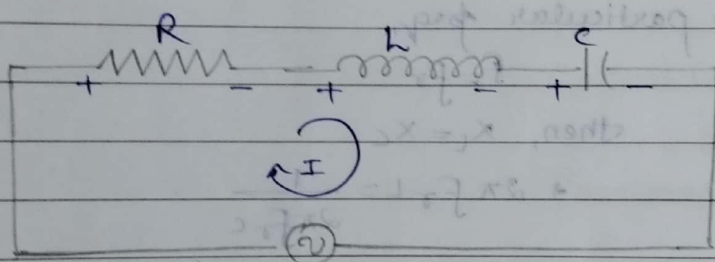
ie, $X_L = X_C$ — condition arises,

the circuit is said to be in resonance.

Resonance is an imp. fn used in many communication circuits, such as a radio receiver.

LCR — Series — Simple called as resonance circuit.

LCR — parallel — anti-resonance circuit or parallel-resonance.



$$v = V_m \sin \omega t$$

Impedance in series R-L-C circuit is —

$$Z = R + j(X_L - X_C)$$

$$= R + j\left(\omega L - \frac{1}{\omega C}\right)$$

Resonant frequency :-

for given R-L-C circuit, one value of

$X_L = X_C$ — only at a particular frequency.

frequency at which $X_L = X_C$ — resonant frequency denoted by f_r .

The series ckt is said to be resonant if one reactive part will become '0'.

ie $Z \neq 0$ $Z = IR$ { Resistive ckt circuit }.

Now, if V across $RLC = \text{const}$, then ϕ freq. rises from 0 gradually, following diff. will be seen,

X_L — inductive reactance rises gradually.
 $= 2\pi fL$

capacitive reactance,

$$X_C = \frac{1}{2\pi fC}$$

Let, at particular freq.,

$$f = f_r$$

then, $X_L = X_C$

$$\rightarrow 2\pi f_r L = \frac{1}{2\pi f_r C}$$

$$4\pi^2 f_r^2 LC = 1$$

$$f_r^2 = \frac{1}{4\pi^2 LC}$$

$$f_r = \frac{1}{2\pi \sqrt{LC}} \quad \text{--- resonant frequency Hz}$$

$$\omega_r = \frac{1}{\sqrt{LC}} \quad \text{--- rad/sec.}$$

$\therefore$ At resonance freq. (f_r)

$$X_L = X_C$$

$$\text{ie } V_L - V_C = 0.$$

RLC will act as purely resistive ckt.

classmate

DATE

Impedance below resonance —

ie $(f < f_r)$

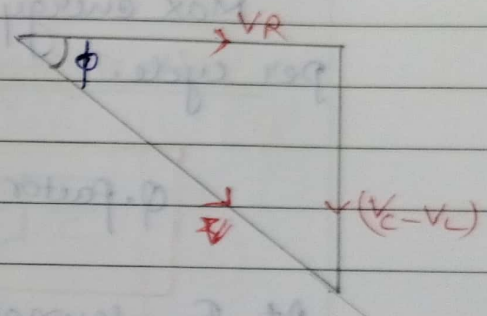
→ more inductive reactance

$X_L < X_C$ (capacitive reactance)

— ie,

$V_L - V_C \rightarrow -ive$

Thus ckt behave as an capacitive ckt. ϕ i-lead voltage by angle ' ϕ '



Impedance above resonance —

ie $f > f_r$

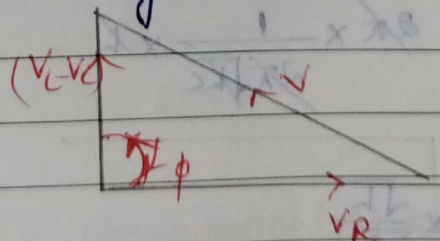
ϕ

inductive reactance > capacitive reactance

$X_L > X_C$

then, $V_L - V_C \rightarrow$ inductive ckt (behaves as an)

i- lags behind 'v' by ϕ



$V_L - V_C \rightarrow +ive$

Poly-phase AC circuits

DATE

--	--	--	--	--	--

Domestic AC Supply is 1- ϕ AC supply of 230V, 50Hz.
But some load need poly-phase AC supply.

Poly phase AC circuit :-

polyphase AC circuit is one which produces many phases simultaneously.

Single generator (armature winding) $\rightarrow$ used to generate 1- ϕ .

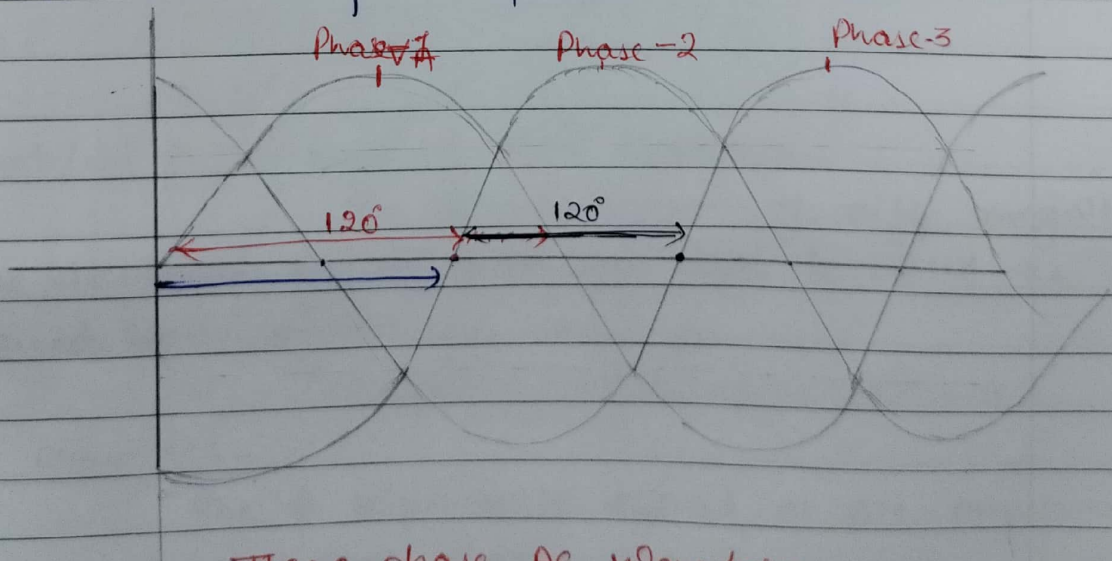
Similarly, more than one arm. winding required to produce ac waveforms.

No. of arm. windings = No. of phases.

There are different no. of poly phase system,
most economically used is - 3- ϕ AC system.

Three-phase power was first adopted by 'Nikola Tesla'. He proved, 3- ϕ is more efficient, transmit & easy to consume electricity.

In rare cases six phase is also used.



Three-phase AC waveform

All the wave forms have identical amplitude & frequency, but they are displaced by 120° .

Advantages of Polyphase System Over 1- ϕ System:-

The poly phase system has more advantages than over 1- ϕ system. They are as follows:-

1) More o/p :-

For the same size, one output of poly phase is about 45% higher than that of 1- ϕ .

2) Smaller size -

For producing same o/p, one size of 3- ϕ m/c is always smaller than 1- ϕ , which is beneficial for urban area.

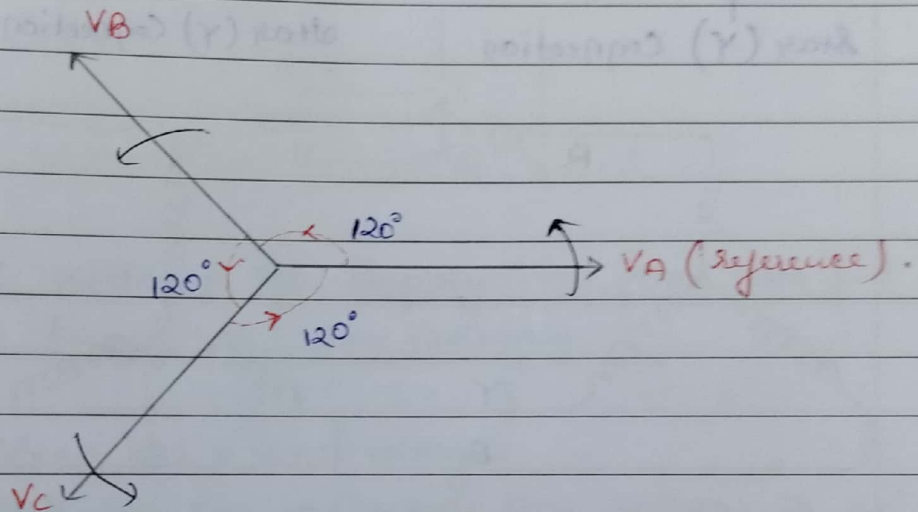
3) 3- ϕ motors are self-starting :-

Mathematical representation of 3- ϕ voltages -

Three phases are identical in magnitude & freq. but they are displaced by 120°

ie -

$$V_A = V_m \sin \omega t$$
$$V_B = V_m \sin(\omega t - 120^\circ)$$
$$V_C = V_m \sin(\omega t - 240^\circ) = V_m \sin(\omega t + 120^\circ)$$



Symmetrical or balanced system:-

The three- ϕ system in which three- ϕ voltages are of identical magnitude & freq. & are displaced by 120° from each other is called a symmetrical or balanced system.

Symmetrical or Unbalanced load system:-

The three- ϕ system of diff. magnitude but are displaced by 120° from each other is called as Unbalanced load system.

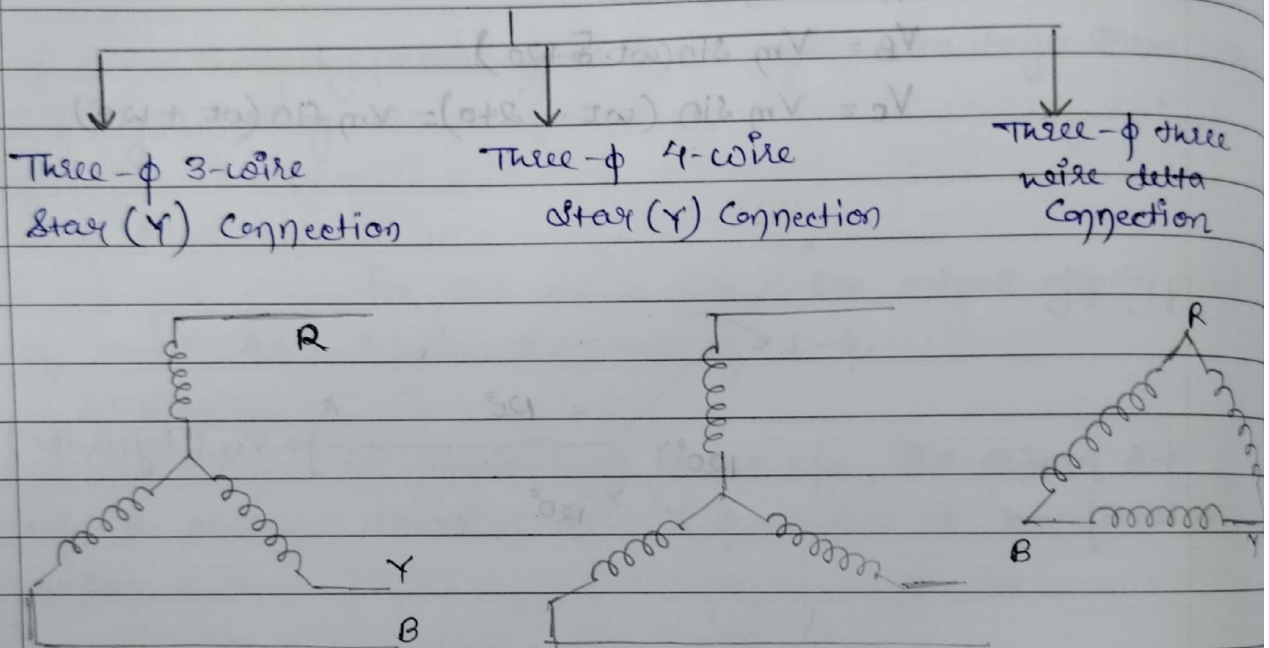
Phase sequence:-

The ϕ -sequence is defined as one sequence in which three- ϕ reach their max positive value.

ie R-Y-B.

Three phase Supply Connection :-

Three- ϕ supply connections are inter-connected in two different ways to reduce one no. of wires is required -



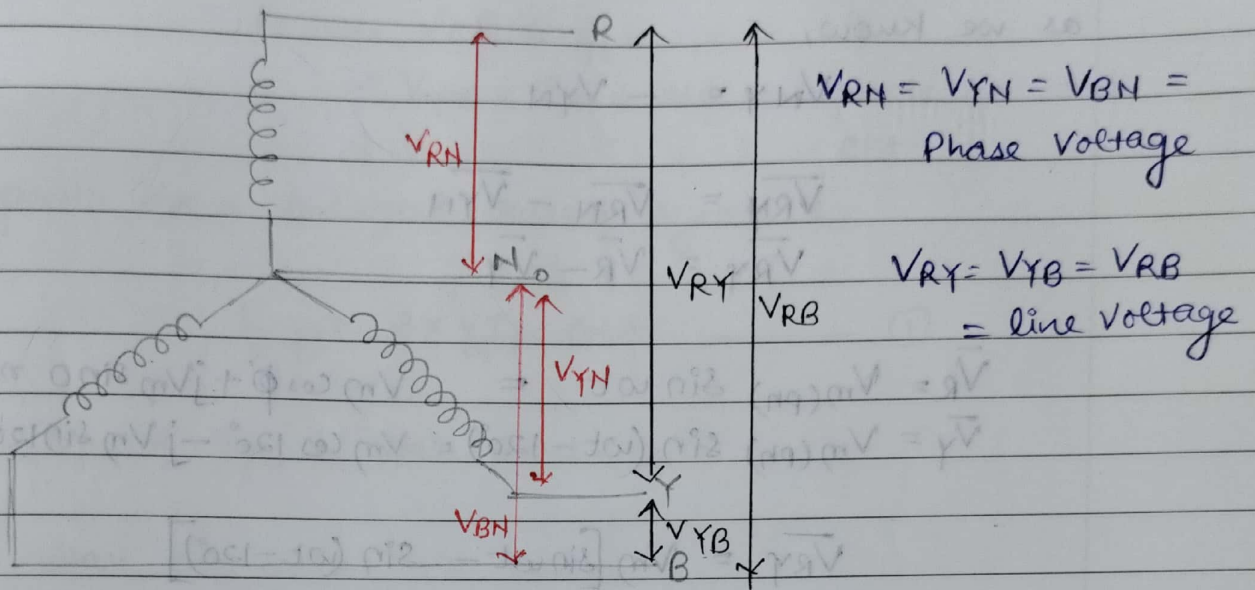
Types of loads :-

On the basis of the winding connection, there are two types of the load.

- 1) Star Connected load
- 2) Delta Connected load.

Star Connected load :-

Star Connected load is also known as Wye Connection.



If R, Y, B are called as three supply lines, then the p.d b/w the lines are known as line voltages.

$V_{RY} = V_{YB} = V_{BR} = \text{line voltages}$
 phase shift b/w the adjacent line voltage is 60° .

The phase voltages are only one voltages measured w.r.t to neutral & having phase difference of 120°

$V_{RN} = V_{YN} = V_{BN} = \text{phase voltage}$

Relation b/w phase Voltage & line Voltage :-

let, $V_{RY} = V_{YB} = V_{BR} = V_L = \text{line voltage}$

$V_{RN} = V_{YN} = V_{BN} = V_{ph} = \text{phase voltage}$

let us study, of RY first,

$$\bar{V}_{RY} = \bar{V}_{RN} + \bar{V}_{NY}$$

as we know,

$$V_{NY} = -V_{YN}$$

$$\bar{V}_{RN} = \bar{V}_{RN} - \bar{V}_{YN}$$

$$\bar{V}_{RY} = \bar{V}_R - \bar{V}_Y$$

$$\bar{V}_R = V_m(\text{ph}) \sin \omega t = V_m \cos \phi + j V_m \sin \phi \Rightarrow V_m \cos 0$$

$$\bar{V}_Y = V_m(\text{ph}) \sin (\omega t - 120^\circ) = V_m \cos 120^\circ - j V_m \sin 120^\circ$$

$$\bar{V}_{RY} = V_m [\sin \omega t - \sin (\omega t - 120^\circ)]$$

$$\bar{V}_{RY} = V_m(\text{ph}) \{ \cos 1 - \cos 120^\circ + j \sin 120^\circ \}$$

$$\Rightarrow V_m(\text{ph}) \{ 1 - (-0.5) + 0.866j \}$$

$$= V_m(\text{ph}) \{ 1.5 + j 0.866 \} \rightarrow \text{converting into polar form.}$$

$$V_L = V_m(\text{ph}) \sqrt{(1.5)^2 + (0.866)^2} \angle \tan^{-1} \left(\frac{0.866}{1.5} \right)$$

$$V_L \rightarrow \sqrt{3} V_{ph} \angle 30^\circ$$

$$\therefore \boxed{\text{Line Voltage} = \sqrt{3} \times \text{phase Voltage}}$$

Relation b/w line current & phase current :-

As the current flowing in each line is equals to the current flowing in the corresponding branch the

$$\boxed{\text{line current} = \text{phase current.}}$$

$$I_L = I_{ph}$$

Power relation :-

$$\begin{aligned} \text{Power} &= V \times I \cos \phi \\ &= V_{ph} \times I_{ph} \times \cos \phi \quad \text{--- power for 1-}\phi \text{ ckt} \end{aligned}$$

$\therefore$ power for a balanced three- ϕ system is,

$$P_{(3-\phi)} = 3 \times V_{ph} I_{ph} \cos \phi \quad \text{--- (1)}$$

V_{ph} & I_{ph} - RMS value.

for star connection we know,

$$V_L = \sqrt{3} V_{ph}$$

$$\& I_L = I_{ph}$$

Substitute one value of V_{ph} & I_{ph} in eqⁿ (1)

$$P_{(3-\phi)} = \frac{3 \times V_L}{\sqrt{3}} \times I_L \cos \phi$$

$$P = \sqrt{3} V_L I_L \cos \phi \quad \text{--- Active power}$$

1) Apparent power -

$$P = 3 V_{ph} \times I_{ph} \quad \text{--- Phase value}$$

$$= 3 \times \frac{V_L}{\sqrt{3}} \times I_L \quad \rightarrow$$

$$P = \sqrt{3} V_L \times I_L \quad \text{--- line value}$$

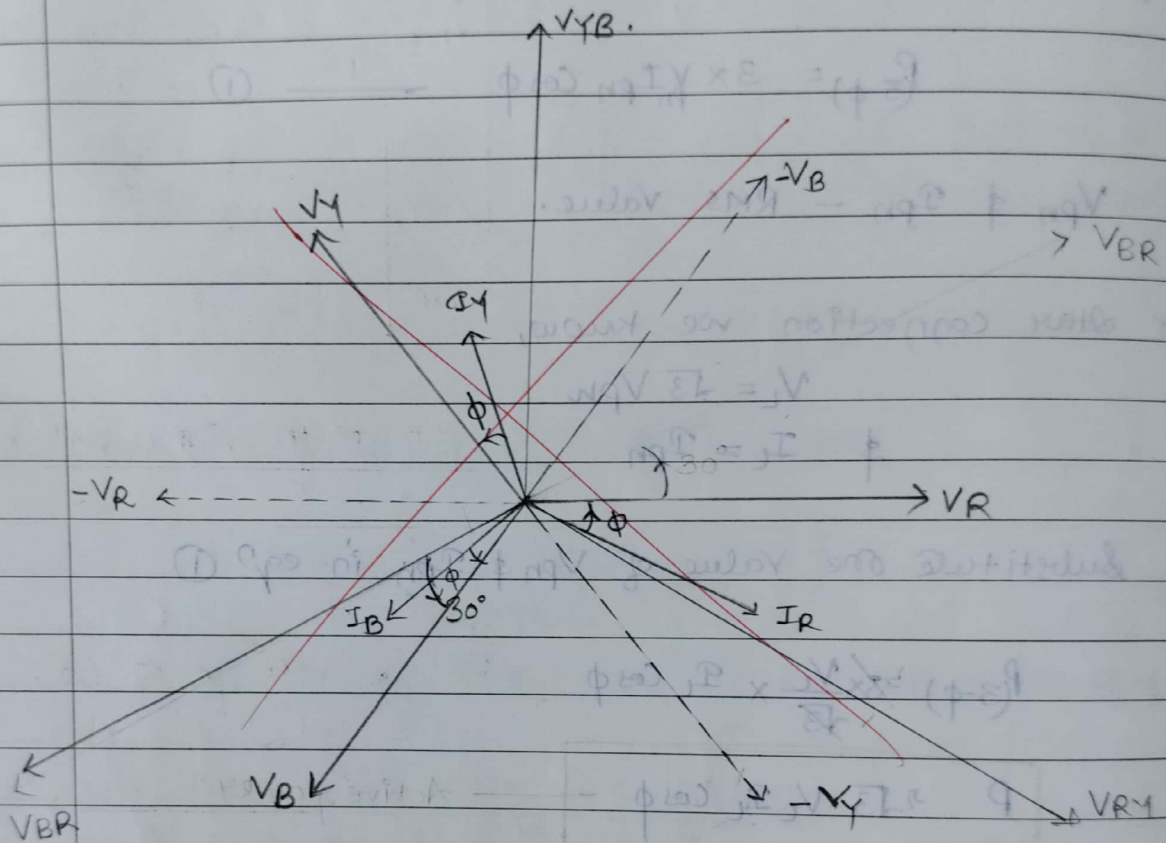
2) Active power -

$$\begin{aligned} P &= 3 V_{ph} I_{ph} \cos \phi \\ P &= \sqrt{3} V_L I_L \cos \phi \end{aligned} \quad \left. \begin{array}{l} \\ \end{array} \right\} \text{for star connection}$$

3) Reactive power—

$$P = 3 V_{ph} I_{ph} \sin \phi$$

$$Q = \sqrt{3} V_L \times I_L \sin \phi \text{ VAR.}$$



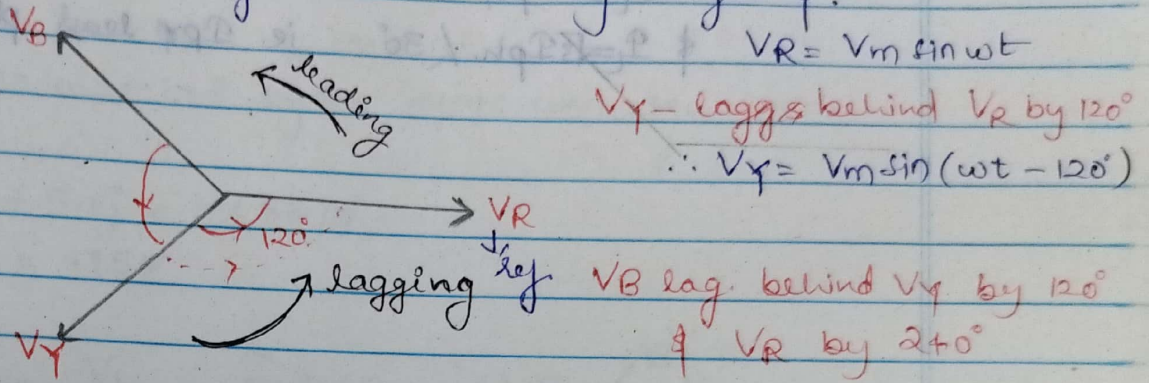
$$V_{RY} = V_{RN} - V_{YN} \\ = V_R - V_Y$$

$$V_{YB} = V_Y - V_B$$

$$V_{BR} = V_B - V_R$$

Phasor diagram of star connected load (inductive nature)

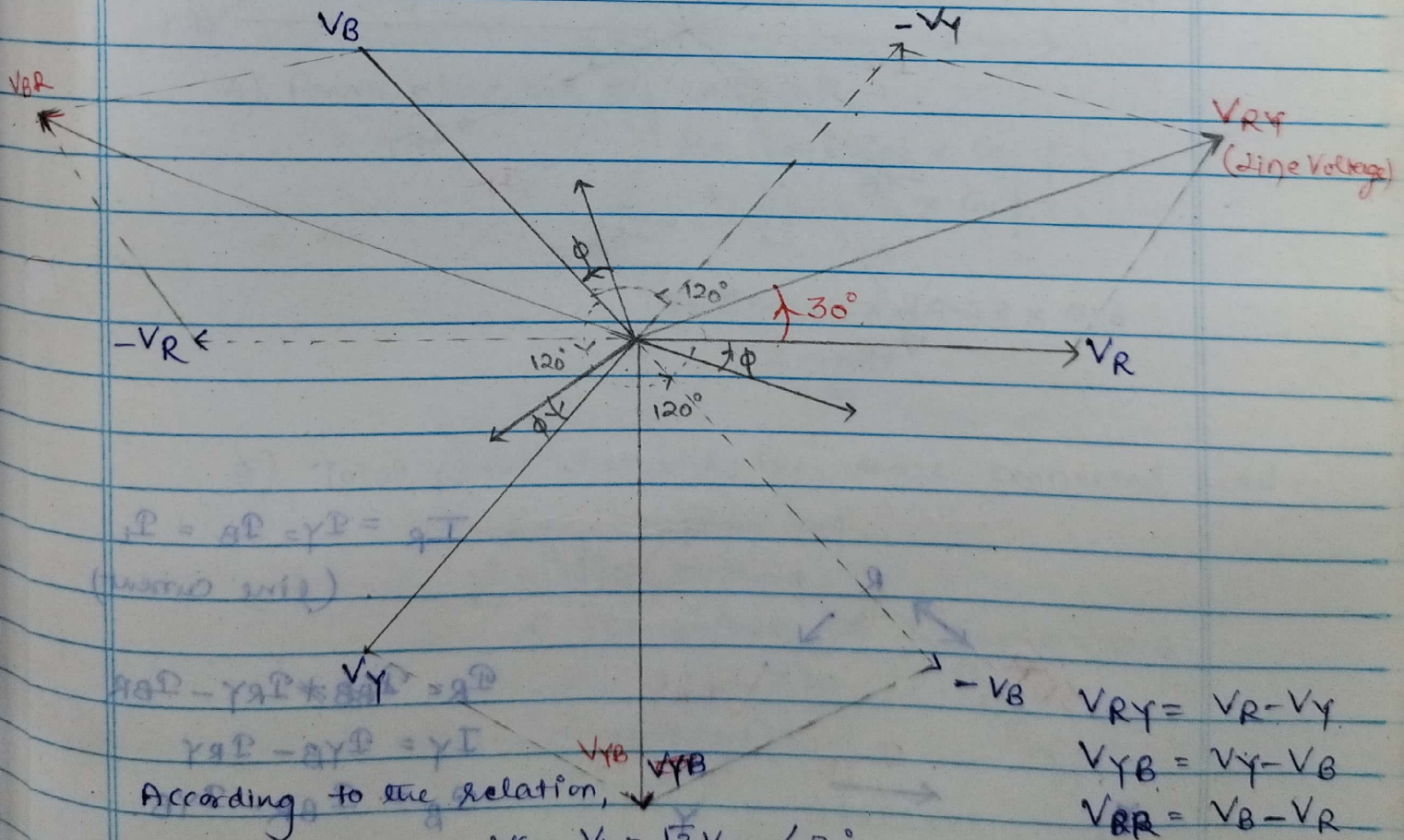
→ In inductive $\rightarrow i$ lags behind v by angle ϕ .



Phasor diagram :-

$$V_B = (V_m \sin \omega t - 240^\circ)$$

or V_B leads V_R by 120°
 $\therefore V_B = V_m \sin(\omega t + 120^\circ)$



According to the relation,

$$V_L = \sqrt{3} V_{ph} \angle 30^\circ$$

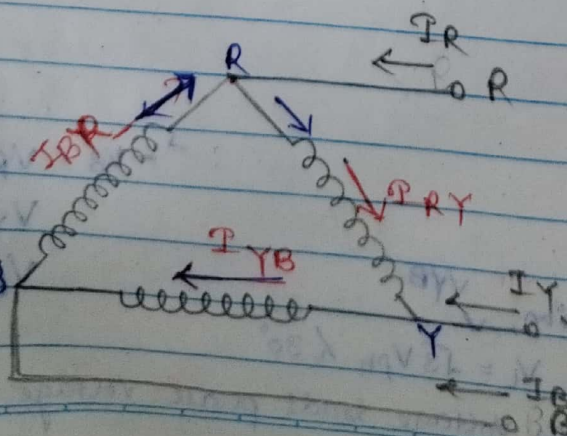
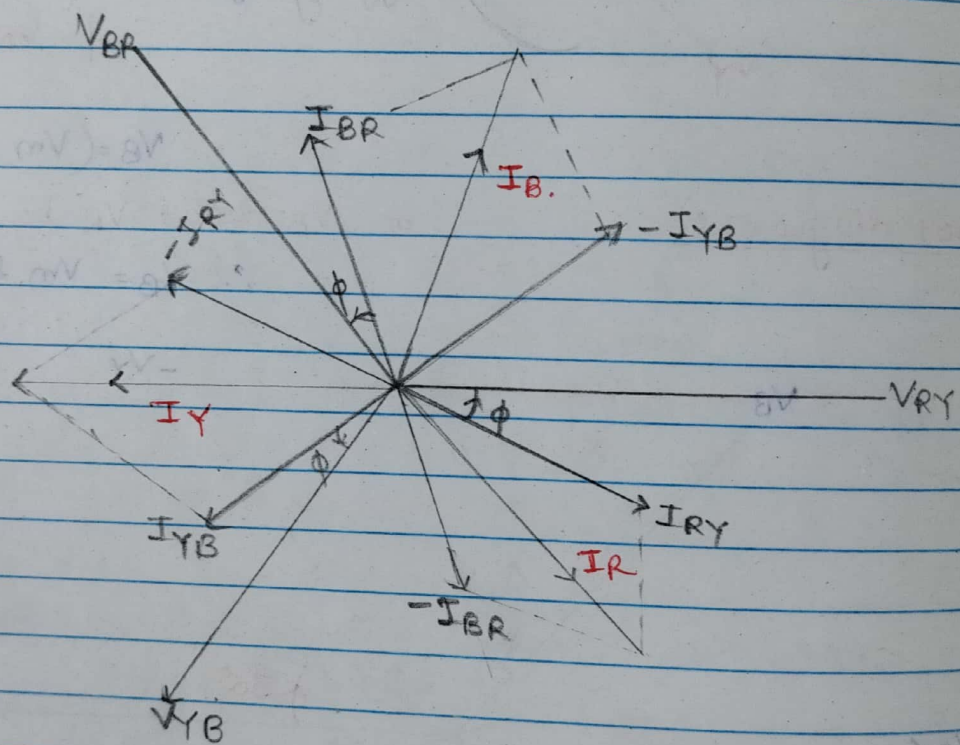
V_{RY} line voltage leads phase voltage by 30° .

Phasor diagram of Delta (Δ) connected load :-

In delta connected load,

$$V_L = V_{ph}$$

$\& I_L = \sqrt{3} I_{ph} \angle 30^\circ$ ie I_{ph} lead phase current I_{RY} by 30°



$$I_R = I_Y = I_B = I_L \quad (\text{line current})$$

$$I_R = I_{YB} - I_{YR}$$

$$I_Y = I_{BR} - I_{RY}$$

$$I_B = I_{RY} - I_{RB}$$

Ques/- Three identical coils each of $(4.2 + j5.6)$ ohm are connected in star across 415V, 3- ϕ , 50Hz supply. Determine,

1. V_{ph} 2. I_{ph} 3. Power factor
4. Power absorbed by each coil
5. power absorbed by one star connected load.

solⁿ. $Z = 4.2 + j5.6 \Rightarrow 7 \angle 53.13^\circ$
 $V_L = 415V$

1) $V_{ph} = \frac{V_L}{\sqrt{3}} \Rightarrow 239.60$ 2) $I_{ph} = \frac{V_{ph}}{Z} \Rightarrow 34.22 \angle -53.13$

$I_L \Rightarrow I_{ph} = 34.23$

3) Power factor ($\cos \phi$) = $\cos(53.13) \approx 0.6$ (lagging)

4) Power absorbed by each coil $\rightarrow$

$$P = V_{ph} \times I_{ph} \times \cos \phi$$

$$\Rightarrow \frac{V_L}{\sqrt{3}} \times I_L \times \cos \phi$$

$$\approx 239.60 \times 34.23 \times 0.6$$

$$\approx \underline{4.9 \text{ Kw}}$$

5) Total power absorbed by star connected load $\rightarrow$

$$P = 3V_{ph} I_{ph} \cos \phi$$

$$\Rightarrow \sqrt{3} V_L I_L \cos \phi$$

$$\approx \sqrt{3} \times 239.60 \times 415 \times 34.23 \times 0.6$$

$$= \underline{14762.7 \text{ W}}$$

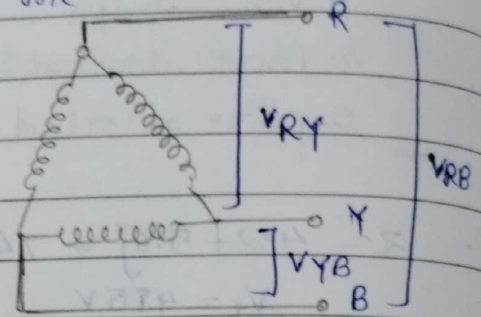
$$\underline{14.7627 \text{ Kw}}$$

Voltage, Current & Power relation in Delta connected load

The line voltage is defined as the p.d b/w two lines,

So, $V_{RY}, V_{YB}, V_{BR},$
 V_{YR}, V_{BR}, V_{RB}

Phase Shift — 60°



The voltage measured across a single winding is called one phase voltage.

V_{RY}, V_{YB}, V_{BR} etc

Typical value of the line voltage & phase for delta connected load is — 440V or 415V

Phase and line voltage in Delta connected load:—

for Delta connection—

$$\text{Line voltage} = \text{phase voltage}$$
$$V_L = V_{ph}$$

Line & phase current relation in Delta connected load:—

Phase current I_{RY} & I_{BR} makes an angle ϕ with V_{RY} & V_{BR} ,

$\therefore I_{RY}$ & I_{BR} having ϕ -displacement of 120°

$$\begin{aligned}\therefore I_{RY} &= I_m(\phi) \sin \omega t = I_m \\ I_{YB} &= I_m(\phi) \sin(\omega t - 120^\circ) = I_m(-0.5 - j0.866) \\ I_{BR} &= I_m(\phi) \sin(\omega t + 120^\circ) = I_m(-0.5 + j0.866)\end{aligned}$$

line current,

$$\begin{aligned}I_R &= I_{RB} + I_{RY} \\ &= -I_{BR} + I_{RY} \\ &= I_{RY} - I_{BR}\end{aligned}$$

$$I_R = I_m - [I_m(-0.5 + j0.866)]$$

$$= I_m + 0.5 I_m - j0.866 I_m$$

$$= 1.5 I_m - j0.866 I_m$$

$$= I_m [1.5 - j0.866]$$

$$I_R = \underline{1.5 I_m} \angle -30^\circ \times I_m(\phi)$$

$$\therefore \boxed{\text{Line current} = \sqrt{3} \text{ phase current}}$$

Power relation $\rightarrow$

1) Total power $\rightarrow$

$$\begin{aligned}P &= 3 V_{ph} \times I_{ph} \\ &= \sqrt{3} V_L \times I_L \quad \text{VA}\end{aligned}$$

2) Active power =

$$\begin{aligned}P &= 3 V_{ph} I_{ph} \cos \phi \\ &= \sqrt{3} V_L I_L \cos \phi \quad \text{W}\end{aligned}$$

3) Reactive power =

$$\begin{aligned}Q &= 3 V_{ph} I_{ph} \sin \phi \\ &= \sqrt{3} V_L I_L \sin \phi \quad \text{VAR.}\end{aligned}$$

Que/- A delta connected balanced load is connected to a 3- ϕ , 400V supply. The load p.f is 0.8 lagging. The current is 34.64 A. find.

1. Resistance, reactance & impedance of one load/phase
2. Total power
3. Total reactive VA

Sol/- 3- ϕ , 400V p.f = 0.8 lagging (inductive)
 $I = 34.64$ A. (line current)

$$Z = |R \cos \phi| + j \sin \phi$$

$$1) \quad Z = \frac{V_{ph}}{I_{ph}} = \frac{400}{I/\sqrt{3}} = \frac{400}{34.64/\sqrt{3}} \approx 20.00 \text{ Amp}^2$$

$$\cos \phi = 0.8 \quad R = Z \cos \phi = 20 \times \frac{0.8}{1} = 16 \Omega$$

$$\therefore X_L = Z \sin \phi = 20 \times \frac{0.6}{1} = 12 \Omega$$

$$Z = R + jX_L = 16 + 12j \Omega$$

2) Total power $\rightarrow$

$$S = \sqrt{3} V_L I_L$$

$$= \sqrt{3} \times 400 \times 34.64 = 23.99 \text{ } 23999.29 \text{ VA}$$

$$P = \sqrt{3} V_L I_L \cos \phi$$

$$= \sqrt{3} \times 400 \times 34.64 \times 0.8 = 19199.43$$

$$= 19.199 \text{ kW}$$

3) Reactive power $\rightarrow \sqrt{3} V_L I_L \sin \phi$

$$Q = \sqrt{3} \times 400 \times 34.64 \times 0.6 = 14399.57 \approx 14.399 \text{ kVA}$$

classmate